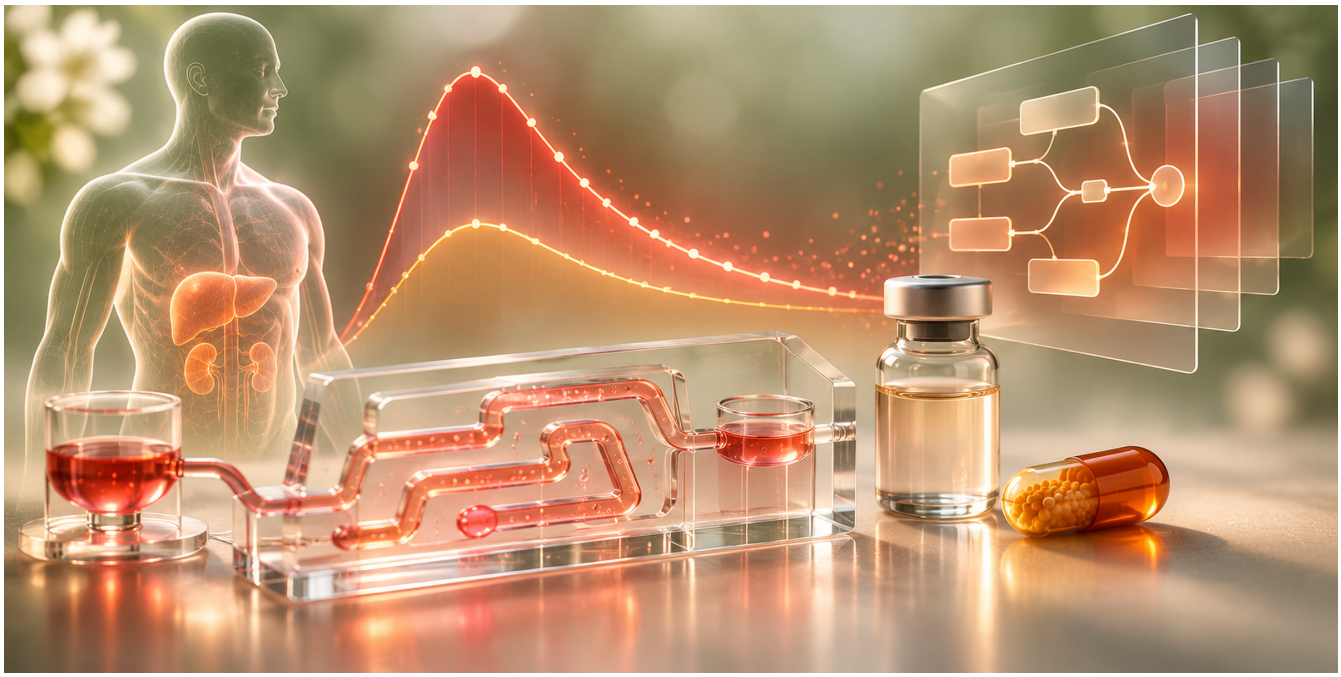
PACEVAL | ORIGINAL-SDK VERIFIED SCIENTIFIC RESEARCH

Patient-Specific Pharmacokinetic Rate Kernels

Reusable and version-controlled mathematical execution for synthetic concentration profiles, PK exposure, sensitivity analysis and numerical model assessment



Concept illustration; not a computed research result. All subjects and compounds in this report are fictional.

Research outcome

94 actual paceval graphs executed on **168 synthetic scenarios** (42 virtual profiles, two models and two administration routes); **417** real multi-graph batch calls producing **1,537,542** values; **9,932** independent 80-digit checks.

Scope: closed-form exact transitions of a one-compartment linear model, not a validated drug, patient data, prescribing instrument or clinical treatment recommendation.

Independent numerical research demonstration | 13 November 2026

1. Introduction: pharmacokinetic use case and state of the art

Drug-development research frequently asks what concentration–time curve an experimental compound might produce under a defined administration schedule, and how that curve would change if clearance, distribution volume or absorption differed between individuals. Pharmacometric teams use established structural and statistical models to investigate these questions. The usual inputs include compound-specific parameters, administration history, physiological covariates, observed concentrations and an uncertainty model. Common outputs include exposure (AUC), sampled peak or trough concentration, model residuals and parameter sensitivity. None of these outputs alone establishes safe or effective clinical dosing.

Current workflows include nonlinear mixed-effects modelling (NONMEM, Monolix, Pumas, nlmixr2), ODE simulation (e.g. rxode2) and physiologically based tools such as PK-Sim/MoBi. Population models require external evaluation, relevant covariate definitions and scientifically qualified data. In 2024, model-ensembling work illustrated the importance of candidate model selection [4]; a 2024/25 clinical pharmacology review documented established PK software [3]. Open Systems Pharmacology offers a substantially broader PBPK environment [7].

Current standards and scientific developments

The ICH M15 framework was adopted at Step 4 on 29 January 2026. The FDA issued final guidance in June 2026, and the EMA lists EU effectiveness from 23 July 2026 [1, 2]. This guideline addresses the planning, evaluation and reporting of model-informed drug-development evidence. It does not authorize an unvalidated mathematical kernel as a medical product. FDA guidance on population PK from February 2022 [5] and more recent literature on external model evaluation [6] reinforce the distinction between numerical execution and clinical qualification.

Research opportunity: reliable reuse of mathematical expressions

The narrowly defined experiment evaluates whether relevant rate, exposure, cost, constraint and sensitivity functions can be assembled into reusable original-paceval computation graphs, then executed for deterministic virtual populations. Compared with a full clinical platform, the project isolates pure mathematical execution; there is no fitting of patient parameters, Bayesian estimation, drug-specific calibration or adaptive-dosing decision logic.

Illustrative input	Mathematical calculation	Research output
Weight, synthetic renal factor, profile perturbations	Two covariate formula versions, CL/V and absorption rates	Independent clearance and half-life versions
Fictional administration events	Exact oral-depot / central-compartment transition	Concentration trace, AUC and sampled peak
Virtual observation / numerical references	Complete residual, 30 weighted-objective graphs and guard margins	Versioned scores and sensitivities

2. Research question and execution architecture

Research question. Can an original paceval Mathematical Engine execute the principal full mathematical chain in a synthetic pharmacokinetic workflow (not merely a trivial rate subterm), preserve two active formula versions, and produce independently verifiable concentration and exposure results?

The C++ host creates the virtual cohort, applies dosing-event state changes, preserves central/depot amounts between intervals, invokes the original batch API, accumulates paceval-derived exposure increments, selects sampled peaks, and writes the output files. paceval owns the scientifically material pure mathematics: clearance, volume, rate constants, exact state transitions, AUC increments, margin expressions, complete objective variants and explicit analytical sensitivities.

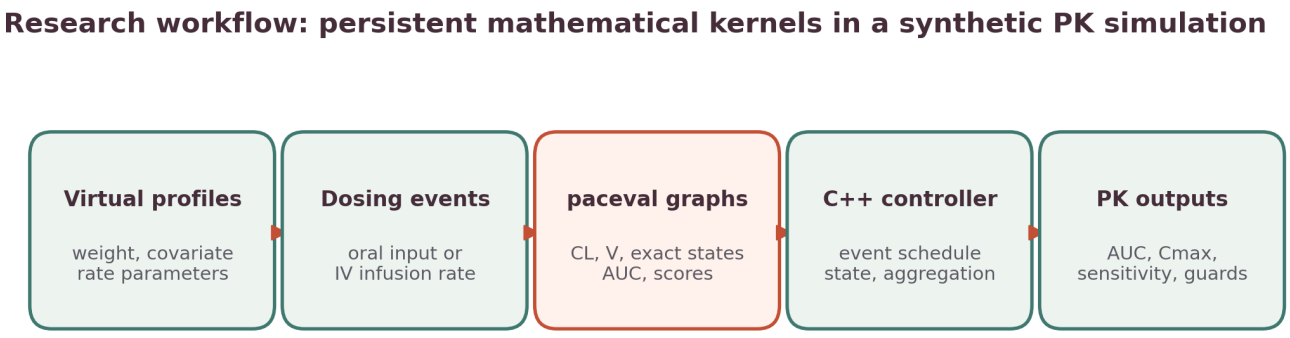


Figure 1. Research architecture. A schematic of the real execution components, not a plot of clinical observations.

What the original SDK actually does

94 distinct formulas are created once using the original **paceval_CreateComputation**, then evaluated repeatedly using **paceval_dGetMultipleComputationsResultsExt**. The program first tests a mixed-function/mixed-row buffer ordering with scalar **paceval_dGetComputationResult** calls. The canonical release records **417** multi-graph batches and **1014** scalar control evaluations.

Every graph is listed in **results/math_execution_inventory.csv** with its mathematical expression, units, version, owner and verification path. No mathematical result in the figure set is produced by running a reference model in place of paceval.

3. Mathematical model, dimensions and assumptions

The source WP_046 proposed the one-compartment concentration RHS dC/dt = R/V − CL·C/V. To cover absorption, event-driven dosing and model changes, the experiment adds a first-order depot and derives an exact state transition for piecewise constant parameters. There are two simultaneously retained hypothetical clearance versions for each synthetic subject.

Covariate parameters; intentionally fictional

$$CL_1 = CL_0(WT/70)^{0.75} RF\, e^{\eta_{CL}}$$

$$CL_2 = CL_0(WT/70)^{0.75}(0.55RF + 0.45RF^2)e^{\eta_{CL}}$$

$$V = V_0(WT/70)e^{\eta_V}, \qquad k = CL/V$$

WT is virtual body weight in kg; CL and CL0 are in L/h; V and V0 in L; k in h^{-1}; and RF is a **dimensionless experimental parameter**, not actual eGFR or a clinical renal-function estimate. The 0.75 allometric exponent is a familiar modelling convention, but the v2 polynomial and all numerical coefficients are unestimated synthetic research choices [3, 8].

Rate equations and analytical step

$$\dot{A}_g = -k_a A_g, \quad \dot{A}_c = F k_a A_g + R - kA_c, \quad C = A_c/V$$

$$A_g' = A_g e^{-k_a h}$$

$$A_c' = A_c e^{-kh} + \frac{F k_a A_g}{k_a - k}(e^{-kh} - e^{-k_a h}) + \frac{R}{k}(1 - e^{-kh})$$

$$\Delta AUC = \frac{F(A_g - A_g') + Rh - (A_c' - A_c)}{CL}$$

Here Ag and Ac are depot and central amounts (mg), ka is absorption rate (h^{-1}), F is the oral bioavailable fraction, R the active infusion rate (mg/h), h the step interval (h), C in mg/L, and AUC in mg·h/L. The final expression is obtained from the exact elimination mass balance. A new administration can change the depot amount or R only at the step boundary. The tested domain excludes ka near k, where the displayed analytical form requires a stable limiting expression.

4. Verified mathematical execution coverage

The SDK has 94 executed persistent graphs: 46 functions for each of two versions, plus two cross-version delta graphs. Each version includes the core compartment and covariate formulas, analytical exposure readouts, single-dose controls, explicit sensitivity functions and fifteen independently evaluated objective variants. The synthetically defined observations and reference margins are numerical test variables, not real measured drug concentrations or therapeutic cut-offs.

Function / domain purpose	Actual owner	Examples of registered graph IDs
CL, V, k, half-life; covariate transforms	paceval	clearance_v1/v2; volume_v1/v2; elimination_constant_v1/v2
Oral and IV fluxes; full state transitions; concentration	paceval	gut_rate; central_rate; depot_next; central_next; concentration_next
Analytical exposure and single-dose PK extrema	paceval	auc_exact_step; auc_infinity_oral; single_oral_tmax; single_oral_cmax
Scores and numerical guard margins	paceval	score_v1_01 ... score_v2_15; upper_margin; relative_clearance_margin
Sensitivities and model-revision deltas	paceval	cl_sensitivity_rf; concentration_next_sensitivity_rf; clearance_delta_v2_v1
Event times, retained state and cross-time reductions	C++ host	Daily event handling, AUC reduction, sampled max selection
Scientific oracle and trajectory audit	Independent Python/C++	verify_results.py and native C++ reference functions

Independent reference and traceability

The executable performed **2286** independent native C++ checks; the separate 80-decimal-digit mpmath oracle verified **9776** selected actual graph-probe outputs and **156** whole-trajectory quantities across **12** scenarios. All 94 graph families are covered. The audit passed. Full numeric verification and tolerances are retained in machine-readable technical files, not reproduced as competing-accuracy tables in this scientific report.

The state solver uses a closed-form analytical transition, **not** an adaptive ODE algorithm; adaptive-step behaviour requested in the original proposal is outside the scope of these reported experiments. TINC was not tested, and no general global error or clinical validity guarantee is asserted.

5. Deterministic virtual-population experiment

The fixed workload comprises seven virtual weights (50–110 kg), six dimensionless synthetic renal multipliers, deterministic small covariate heterogeneity terms, two simultaneously available clearance versions and two distinct routes: 240 mg oral into a depot at 24-hour intervals, or 120 mg/h intravenous infusion for the first two hours each day. Four days contain 384 quarter-hour steps and 168 version × route × profile combinations. Bioavailability is below unity for the oral scenarios, so the alternative routes do not represent identical systemic input.

Research input	Value / interpretation
Reference parameters	CL0 = 5 L/h; V0 = 40 L; profile-specific ka and bioavailability
Observation / score controls	3 mg/L model target; 6 mg/L illustrative numerical ceiling; AUC reference 40 mg·h/L
Time horizon	0–96 h, h = 0.25 h, oral/IV events each 24 h
Model revisions	v1: RF; v2: 0.55 RF + 0.45 RF² in clearance
Published outputs	Scenario summary, actual concentration traces, graph registry and probe inputs

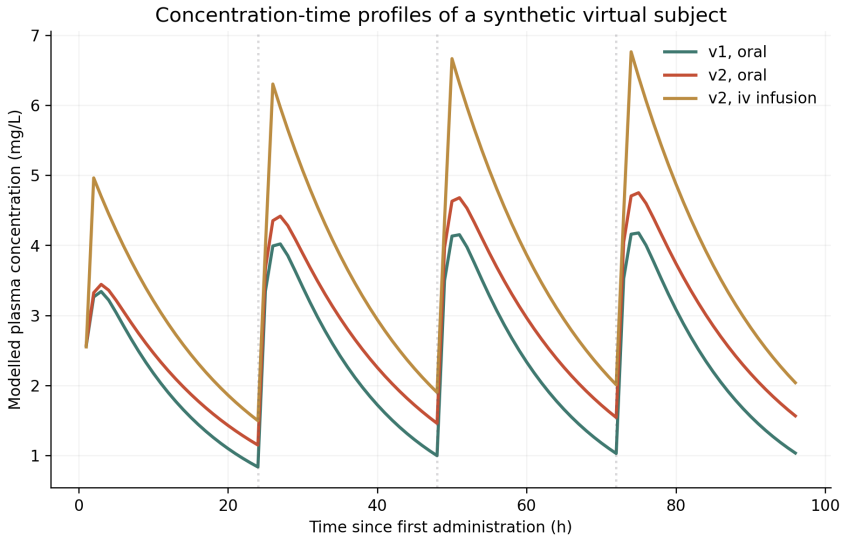


Figure 2. Versioned model trajectories and an intravenous alternative for fictional profile 19. All plotted values come from the executed original paceval graphs.

The sampled Cmax reduces the 0.25-hour concentration grid, while the AUC sums exact per-step paceval exposure expressions. The two different summary methods must not be confused with continuous-time peak finding or exposure extrapolated beyond 96 hours.

6. Modelled exposure: renal factor and body weight

Across 42 synthetic oral profiles, the median 96-hour AUC is **148.464 mg·h/L** for version 1 and **156.215 mg·h/L** for version 2. These differences arise solely from fictional covariate formulas and synthetic subjects, not clinical evidence.

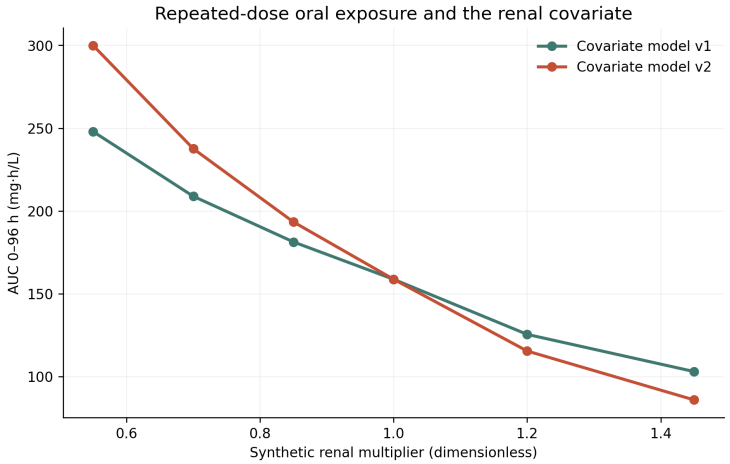


Figure 3. For 70-kg virtual subjects, AUC(0–96 h) changes systematically across the dimensionless renal multiplier.

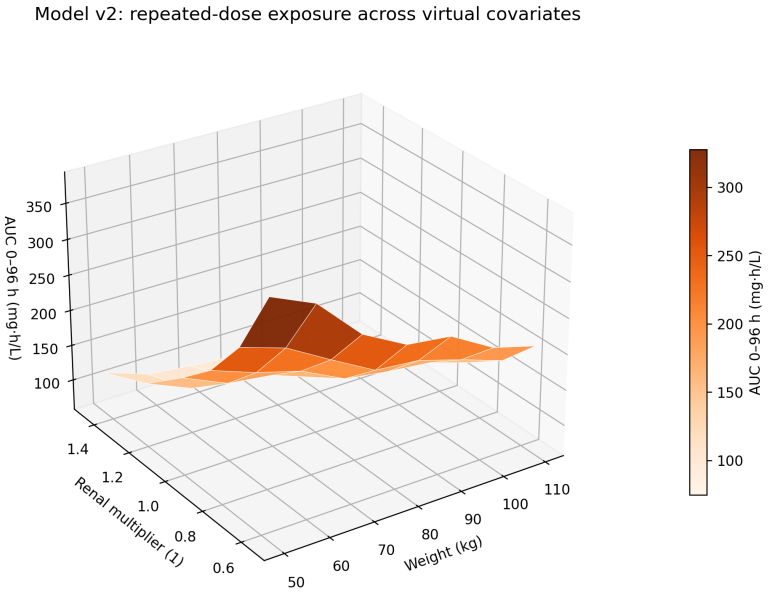


Figure 4. Three-dimensional surface of 42 measured original-paceval output rows (v2, oral); weight and renal factor jointly vary.

7. Controlled formula revision and peak response

The v2 quadratic covariate family is a retained model revision: both versions are available without replacing or mutating previously registered graph handles. The AUC difference plotted below is formed from actual scenario outputs under the same profile and administration schedule. It is not evidence that one formula is pharmacologically preferable.

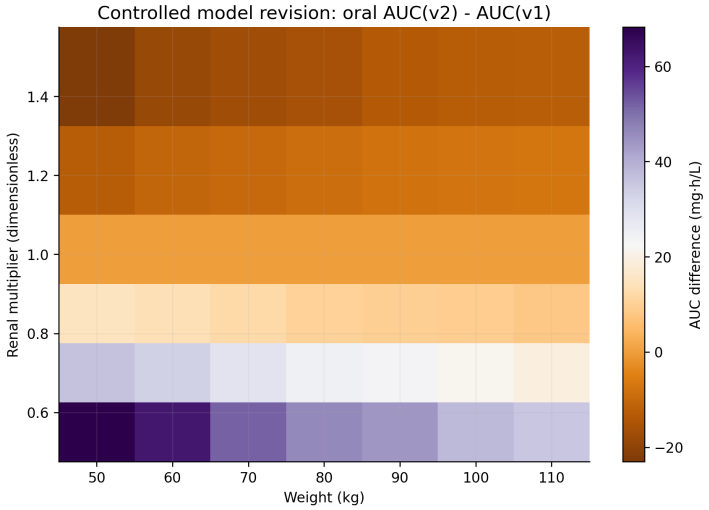


Figure 5. AUC(v2) minus AUC(v1) for the 42 fixed oral profiles, in mg·h/L.

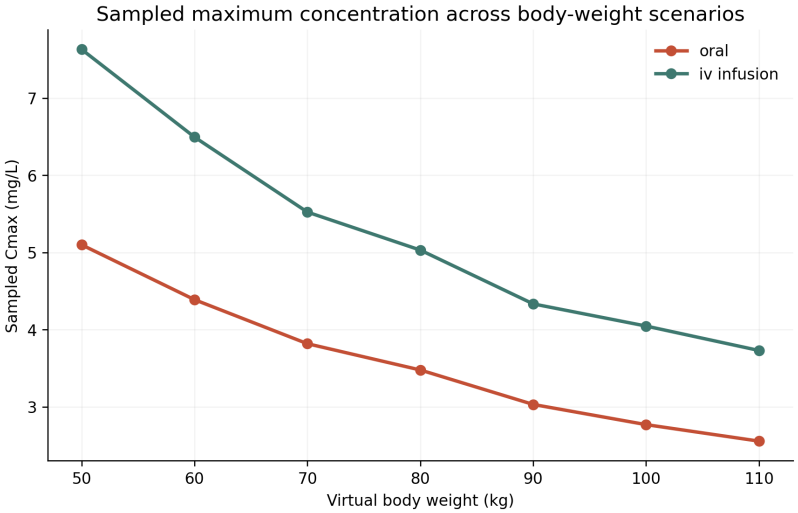


Figure 6. Cmax sampled at 0.25 h resolution for virtual weight variation under v2. Oral and IV systemic input are not matched.

8. Distinct subject-level responses and PK summary metrics

Under v2, median sampled oral peak concentration across 42 profiles is **3.671 mg/L** versus **5.367 mg/L** for the illustrative IV regimen. These are numerically defined scenarios of a fictional molecule, not a comparative recommendation for routes of administration.

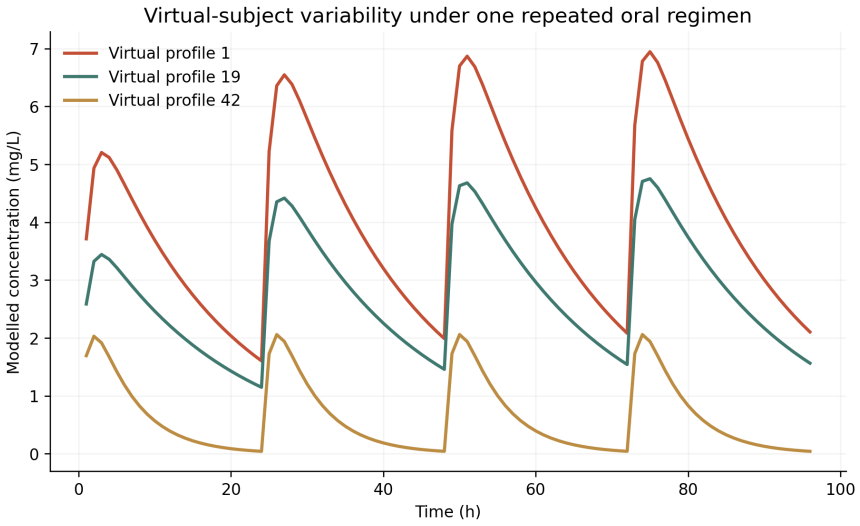


Figure 7. Three representative concentration time courses vary because the virtual profiles differ in their deterministic parameters.

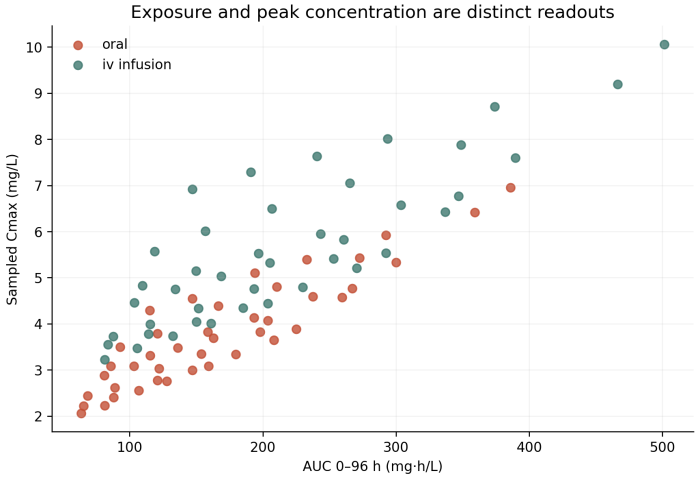


Figure 8. AUC and sampled Cmax carry complementary information and are not interchangeable summaries.

9. Parameter sensitivities and numerical guard functions

The same real graph inventory contains explicit derivatives of clearance with respect to synthetic RF and weight, volume with respect to weight, oral AUC with respect to RF, and one-step concentration and squared-residual objectives with respect to RF. These are hand-derived mathematical expressions verified independently; no undocumented automatic differentiation is claimed.

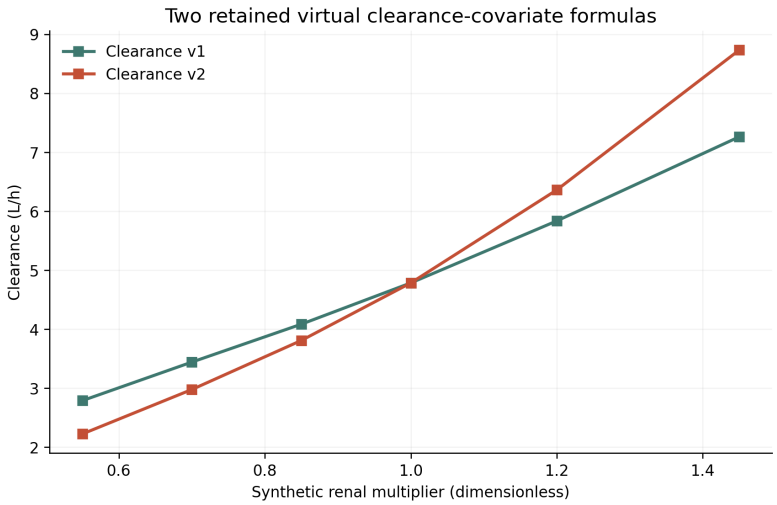


Figure 9. Version-preserving clearance formulas for virtual 70-kg subjects. The renal multiplier is dimensionless.

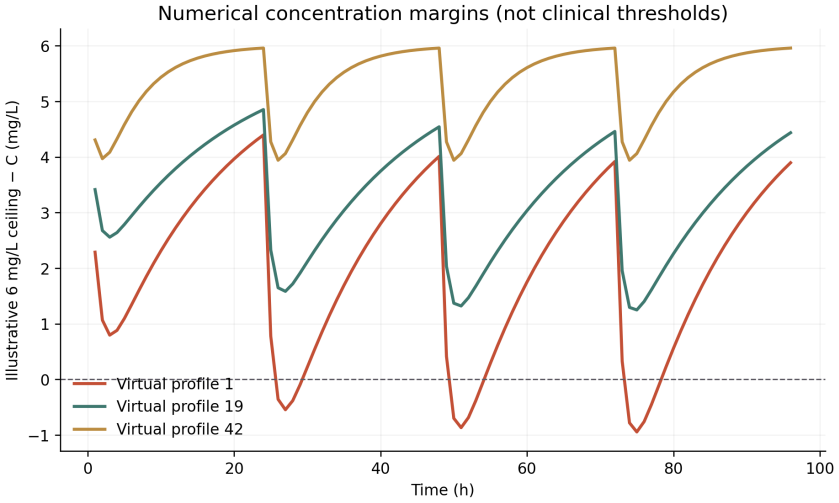


Figure 10. Distance to an arbitrary 6 mg/L numerical ceiling, computed by paceval. Negative values indicate a violated illustrative inequality, not a clinical safety assessment.

10. Interpretation, verification and scientific limitations

A fully traceable synthetic result

For virtual profile 19 under the four-day oral schedule, original paceval results are AUC(0–96 h) **220.853682** and **267.203110 mg·h/L** in v1 and v2 respectively. The corresponding sampled Cmax values are **4.208479** and **4.768603 mg/L**. The retained input vector, graph expressions, output files and independent high-precision oracle permit reproduction and audit. This is a mathematical demonstration, not an exposure target.

Outcome for synthetic profile 19, oral	Model v1	Model v2
CL (L/h)	3.13265	2.49829
AUC(0–96 h) (mg·h/L)	220.85368	267.20311
Sampled Cmax (mg/L)	4.20848	4.76860
End-of-day-4 trough (mg/L)	1.03329	1.56382

Verified scope

Actual original SDK graph creation and repeated multi-graph batch evaluation; distinct simultaneous formula versions; full exact state transition, exposure and composite-score graph computations; independent native and 80-digit references; defined units, domain constraints, deterministic event handling and documented scientific chart provenance. The CMake project and CTest chain are supplied.

Boundary between mathematics and medicine

All subjects and drug parameters are synthetic, with no clinically fitted renal relationship, drug-specific assay, real observation record, measured PK residual distribution, structural model validation, identifiability analysis, external predictive assessment or validated therapeutic window. The case excludes ka approximately equal to k and does not investigate nonlinear elimination, multiple compartments, Bayesian model fitting, population inference, adaptive integration, clinical safety, regulatory clearance or TINC. Finite checks do not prove a universal numerical accuracy property.

Future work could add properly licensed, anonymized observations and independently accepted drug-specific models, with external data qualification and expert review. Such studies should be treated as new research projects, not as an inference already justified by these outputs.

11. Reproducibility and external scientific sources

The ZIP contains the original paceval SDK wrapper files under src/ (upstream copyright and AGPL statements preserved), the CMake executable, deterministic virtual-profile data, machine-readable execution summaries and graph registry, independent mpmath verifier, figure regeneration code and this PDF generator. The externally supplied paceval Linux runtime is excluded and must be provided separately in a manner consistent with its license.

Build: cmake -S src -B build -DCMAKE_BUILD_TYPE=Release -DPACEVAL_RUNTIME=/absolute/path/to/libpaceval_linux_sharedLIB.so

Run: cmake --build build -j ; ctest --test-dir build --output-on-failure ; ./build/pharmacokinetic_kernels --out . --steps 384 --patients 42

Independent checks and graphics: python3 src/verify_results.py --root . ; python3 src/make_figures.py --root . --figures figures ; python3 src/check_readme_math.py ; python3 src/make_whitepaper.py --root .

Artifact path	Function
results/math_execution_inventory.csv; formula_registry.csv	Exact graph formulas, versions, units, ownership and purpose
results/graph_probe.csv; sdk_execution.txt	Real batch-input vectors and computed outputs; actual execution log
results/scenario_summary.csv; concentration_trajectories.csv	Full scientific application results for all 168 scenarios / selected traces
results/high_precision_audit.json; verification_summary.json	Independent 80-decimal-digit whole-result and graph validation status
results/figure_manifest.csv; src/make_figures.py	Figure sources, captions, fully regenerated research graphics

Scientific and technology references

[1] ICH, M15 General Principles for Model-Informed Drug Development. Step 4, January 2026; EMA effective July 2026: ema.europa.eu/en/ich-m15-guideline-general-principles-model-informed-drug-development-step-2b-scientific-guideline. [2] FDA, M15 final guidance, June 2026: fda.gov/regulatory-information/search-fda-guidance-documents/m15-general-principles-model-informed-drug-development.

[3] Recommended approaches for integration of population pharmacokinetic modelling with precision dosing in clinical practice, British Journal of Clinical Pharmacology, DOI 10.1111/bcp.16335. [4] Selecting the Best Pharmacokinetic Models for a Priori Model-Informed Precision Dosing with Model Ensembling, 2024, PMC11522197. [5] FDA, Population Pharmacokinetics: Guidance for Industry, February 2022.

[6] Guidance for External Evaluation and Selection of Population Pharmacokinetic Models for Precision Dosing, PubMed 41343019. [7] Open Systems Pharmacology, PK-Sim & MoBi, github.com/Open-Systems-Pharmacology/PK-Sim. [8] Monolix PKPD model library, one-compartment linear absorption/infusion equations. [9] paceval original API and repository: paceval.com/api/ and github.com/paceval/paceval.

Conclusion. Within a clearly delimited synthetic one-compartment workload, this original-SDK study verifies complete reusable pharmacokinetic mathematical graph chains, versioned formula evaluation and independently reproducible PK outputs. It does not establish clinical dosing validity.